FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

# Obstetric & Gynaecological History and Examination

Clinical skill | Week 1 · Foundations, ANC & early pregnancy | Dr Mutamba, Dr Nanzira, Dr Kaduyu

A structured obstetric history (biodata, gravidity/parity, dates, presenting complaint, history of present pregnancy, past obstetric, gynaecological, medical, drug, family and social history) plus a general, obstetric abdominal (inspection, fundal height/SFH, Leopold manoeuvres, lie, presentation, position, engagement in fifths, FHR), speculum and bimanual examination lets you find risk, make a diagnosis and present a crisp summary on a ward round.

**Must know before the ward round**

1. Gravida is the number of pregnancies ever, including the current one. Para is the number of deliveries at or after 28 weeks in Uganda (24 weeks in the UK), alive or dead. The "+" number is pregnancies lost before viability. Twins count as one para.
2. Always calculate the gestational age from the LNMP (Naegele's rule) and compare it with an early ultrasound and the fundal height. Say which one you trust and why.
3. Fundal height landmarks are 12 weeks at the symphysis, 20–22 at the umbilicus and 36 at the xiphisternum. From 24 weeks, SFH in cm equals weeks ± 2.
4. Leopold manoeuvres are fundal grip (what is at the fundus), lateral grip (where is the back, and so the lie), Pawlik's grip (the presenting part and whether it is mobile) and the pelvic grip (engagement in fifths, attitude).
5. Lie is the relation of the fetal long axis to the mother's. Presentation is the part over the pelvic inlet. Position is the denominator's relation to the maternal pelvis (e.g. LOA). Engagement is 2/5 or less palpable.
6. Normal FHR is 110–160 bpm. Auscultate with a Pinard or Doppler over the fetal anterior shoulder for a full minute, feeling the maternal pulse at the same time.
7. Never do a digital VE on a woman with antepartum haemorrhage until placenta praevia has been excluded by ultrasound. A gentle speculum exam is allowed.
8. Present as a summary line first ("a 24-year-old G2P1+0 at 34 weeks with…"), then the key positives and negatives, the impression, and the plan.
9. Before clerking, check ABC, vital signs, shock index and FHR. BP 160/110 mmHg or more, shock index 0.9 or more, FHR outside 110–160, cord prolapse or fits mean act and call for help (SBAR) first, clerk later.
10. If dating estimates disagree, trust an early scan (before 20–24 weeks) over the LNMP, and the LNMP over fundal height.
11. A clerking is not complete without the bedside tests. Urine dipstick for every obstetric patient, HemoCue Hb, and a urine hCG for every gynae patient of reproductive age.
12. SFH more than 3 cm off dates, malpresentation at 36 weeks or later, and a 5/5 head at term in a primigravida all need an ultrasound (placenta first, before any VE).


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# 1. Purpose and Indications

The **history and examination** is the core clinical skill of obstetrics, and it is what you will be assessed on in every **bedside teaching session, ward round, long case and OSCE**. In Uganda, where investigations are limited and delays kill, a clinician who takes a **focused, risk-oriented history** and does a **systematic abdominal examination** can pick up most pre-eclampsia, malpresentation, multiple pregnancy, growth restriction and obstructed labour **with bare hands, a tape measure, a BP machine and a Pinard**.

An obstetric history differs from a medical one because it has **two patients**, is **time-dependent** (everything is interpreted against the gestational age), and hinges on the **previous obstetric history**, which is the best predictor of what will happen next.

The one thing never to forget: **start every presentation with the summary line: age, gravidity/parity, gestational age (and how it was determined), and the main problem.** Get that right and the rest follows.

| Examination | Indications |
|---|---|
| **Obstetric history and abdominal exam** | Every ANC contact, admission, labour |
| **Speculum exam** | Bleeding (to see the source and whether the os is open), suspected PROM (pooling of liquor), abnormal discharge, cervical screening, suspected cervical lesions, IUD strings |
| **Digital VE / bimanual** | Labour assessment (every 4 hours in the active phase), gynae assessment (uterine size, adnexal masses, tenderness), before induction (Bishop score), cord prolapse check |
| **Rectal exam** | Suspected 3rd/4th degree tear, after perineal repair, gynae oncology staging |

# 2. Contraindications and Precautions

| Examination | Contraindications / cautions |
|---|---|
| **Obstetric abdominal exam** | None, but avoid prolonged supine positioning after 20 weeks (**aortocaval compression**, so use a left tilt). Be gentle in **suspected abruption or uterine rupture** |
| **Speculum exam** | Refusal; take care in APH (gentle) |
| **Digital VE / bimanual** | **APH until placenta praevia is excluded** (can trigger torrential bleeding); **PPROM without labour** (raises infection risk, so use a sterile speculum only); virgo intacta or no consent (use a rectal or transabdominal alternative) |
| **Rectal exam** | Consent; explain why it is needed |

> **RED FLAG**
> **Do not do a digital VE** in a woman with antepartum bleeding (placenta praevia not yet excluded) or with PPROM who is **not** in labour. Write "VE deferred: APH, awaiting scan" so the next person does not do it either.

Other precautions:

- **Stability before history**: a woman who is bleeding, fitting, shocked or pushing is treated first and clerked later (section 5, Step 0).
- **Consent and chaperone** for every intimate examination; a student examination is a privilege the woman may decline.
- **Infection prevention**: sterile gloves for VE in labour; repeated VEs after rupture of membranes raise infection risk.

# 3. Relevant Anatomy and Normal Values

## Obstetric terminology

| Term | Definition | Example |
|---|---|---|
| **Gravida** | Total number of pregnancies, **including the current one**, regardless of outcome | A woman pregnant for the third time is G3 |
| **Para** | Number of pregnancies that ended in delivery **at or beyond the age of viability** (**28 weeks in Uganda**; 24 weeks in the UK), alive or stillborn. **Twins count as 1** | – |
| **"+" (abortions)** | Pregnancies ending **before viability**: miscarriages, induced abortions, ectopic and molar pregnancies | G4P2+1: pregnant now, 2 deliveries, 1 miscarriage |
| **Primigravida** | Pregnant for the first time | G1P0 |
| **Multigravida** | Pregnant for the second or subsequent time | – |
| **Nullipara / primipara / multipara** | Has never delivered / delivered once / delivered 2 or more times beyond viability | – |
| **Grand multipara** | Has delivered **5 or more** times | Risk of PPH, malpresentation, uterine rupture |
| **TPAL** (US system) | Term, Preterm, Abortions, Living children | G4 P2-0-1-2 |
| **LNMP** | First day of the last normal menstrual period | – |
| **EDD** | LNMP + 9 months + 7 days | – |

> **UGANDA CONTEXT**
> Uganda (MoH and most teaching hospitals) uses **28 weeks** as the threshold of viability for parity and stillbirth definitions, and writes parity as **"Para X+Y"** (e.g. **P2+1**). In the UK, 24 weeks is used. Always state the number of **living children** separately ("P3+1, 2 living children"), because a pregnancy loss or child death is critical information.

**Naegele's rule**: EDD = LNMP **+ 7 days – 3 months + 1 year** (= LNMP + 280 days); add (cycle length – 28) days for long cycles.

## Fetal lie, presentation, position, attitude and engagement

| Term | Definition | Normal |
|---|---|---|
| **Lie** | Relation of the **long axis of the fetus** to the long axis of the uterus/mother | **Longitudinal** (others: transverse, oblique) |

| Term | Definition | Normal |
|---|---|---|
| **Presentation** | The part of the fetus that **lies over the pelvic inlet** | **Cephalic (vertex)** (others: breech, shoulder, face, brow, compound) |
| **Presenting part** | The **lowest part felt on VE** | Vertex |
| **Attitude** | Relation of fetal parts to each other (flexion vs extension of the head) | **Flexed** |
| **Denominator** | The reference point on the presenting part | Occiput (vertex), sacrum (breech), mentum (face), acromion (shoulder) |
| **Position** | Relation of the **denominator** to the **maternal pelvis** (left/right, anterior/transverse/posterior) | **LOA / ROA / OA** (6 main positions per presentation) |
| **Engagement** | The widest transverse diameter (**biparietal**) has passed through the pelvic brim, which means **2/5 or less** of the head is palpable abdominally | In primigravidae, often from about 36–38 weeks; in multigravidae, often only in labour |
| **Station** | Level of the leading bony part relative to the **ischial spines** (−3 to +3) | – |

## Fundal height landmarks

| Landmark | Gestation |
|---|---|
| Just palpable above the symphysis | 12 weeks |
| Midway between the symphysis and umbilicus | 16 weeks |
| At the umbilicus | 20–22 weeks |
| Two fingerbreadths above the umbilicus | about 24 weeks |
| Midway between the umbilicus and xiphisternum | about 28 weeks |
| Just below the xiphisternum | 36 weeks |
| Drops slightly (lightening), especially in primigravidae | 38–40 weeks |

**Symphysis–fundal height (SFH)**: from 24 weeks, **SFH (cm) = gestational age (weeks) ± 2 cm** (between 24 and 36 weeks).

## Engagement in fifths (the "rule of fifths")

The head is about **5 fingerbreadths** wide above the brim. Count how many fingers' width of head remains palpable above the symphysis.

```
Fifths palpable     Description                 Station (approx.)
5/5  ──────────────  Free, mobile above the brim   −3 or higher
4/5  ──────────────  Sinciput & occiput easily felt
3/5  ──────────────  Not engaged (widest part above brim) −2 to −1
2/5  ──────────────  ENGAGED (sinciput felt, occiput barely)  0
1/5  ──────────────  Sinciput just felt              +1
0/5  ──────────────  Head not palpable abdominally  +2 or lower
```

## Normal values

| Item | Normal / threshold |
|---|---|
| FHR | **110–160 bpm**, regular |
| BP | Below 140/90 mmHg (severe 160/110 or more) |

| Item | Normal / threshold |
|---|---|
| Maternal pulse | Below 100 bpm |
| Shock index (HR ÷ SBP) | Below 0.9 (0.9 or more abnormal; 1.4 or more critical) |
| Hb in pregnancy | 11 g/dL or more (severe anaemia below 7 g/dL) |
| Maternal height | 150 cm or more (below 150 cm = CPD risk marker) |
| Twins on auscultation | Two FHRs **10 or more bpm apart** |
| Bishop score | **8 or more** favourable; **6 or less** needs ripening |

# 4. Equipment and Preparation

## Equipment

| Item | Use |
|---|---|
| **BP machine** (correct cuff size) and stethoscope | BP at every contact, heart and chest |
| **Thermometer, watch, pulse oximeter** | Vital signs |
| **Weighing scale and height board** | BMI; height below 150 cm is a risk |
| **Non-elastic tape measure** (cm) | Symphysis–fundal height (SFH) |
| **Pinard fetal stethoscope** or **hand-held Doppler** | FHR |
| **Urine dipstick** | Protein, glucose, nitrites |
| **Gloves, lubricant, good light** | Vaginal examination |
| **Cusco's (bivalve) speculum**; **Sims' speculum** (for prolapse and fistula) | Visualising the cervix and vaginal walls |
| **Swabs, slides, VIA kit (acetic acid)** | High vaginal swab, endocervical swab, cervical screening |
| **Chaperone, screen/curtain, drape** | Dignity and safety |
| **ANC card / Mother–Child Health Passport and partograph** | Records |

## Consent, privacy and chaperone

- **Introduce yourself**, confirm her identity, explain what you will do and why, and get **consent**. Ensure privacy (screen or curtain) and ask about a companion or **chaperone** (always for breast, speculum, VE and rectal examination).
- Ask her to **empty her bladder** before an abdominal examination (a full bladder raises the SFH and makes palpation uncomfortable).

## Positioning and exposure

- **Abdominal examination**: supine with **a pillow under her head**, **arms by her sides**, knees slightly flexed (relaxes the abdominal muscles), and a **left lateral tilt** if she becomes uncomfortable. Expose from the **xiphisternum to the symphysis**, keeping the pubic area covered. Stand on her **right**. Warm your hands.
- **Speculum and VE**: **dorsal (lithotomy) position** with heels together and knees apart, good light; Sims' speculum in the **left lateral** position.

# 5. Procedure Step by Step

### Step 0: Is she stable enough to be clerked? (first 5 minutes)

Before you take a 30-minute history, decide: **is she stable enough to be clerked, or does she need action first?**

1. **Look at her**: conscious and talking? In pain? Pale? Fitting? Bleeding onto the bed? Pushing?
2. **ABC and vital signs**: airway, RR, $SpO_2$, pulse, BP, temperature; **shock index** (HR ÷ SBP).
3. **Obstetric quick checks**: gestation (roughly, from the card or the fundus), **FHR**, contractions, any **bleeding or liquor** on the pad, cord or fetal part at the introitus.
4. **Act and call for help (SBAR) at once if:**

| Finding | Immediate action (clerk later) |
|---|---|
| **BP 160/110 mmHg or more**, or fits | Left lateral, $MgSO_4$ loading, antihypertensive (per senior), see the pre-eclampsia topic |
| **Shock index 0.9 or more**, heavy bleeding | Two large-bore cannulae, bloods for Hb and cross-match, fluids; **no VE** if antepartum bleeding |
| **FHR below 110 or above 160, or absent** | Left lateral, stop oxytocin, check cord, inform the midwife/doctor |
| **Cord prolapse / fetal part at the introitus** | Call for help; knee–chest or head-down position, push the presenting part up, fill the bladder, prepare for CS |
| **Fully dilated and pushing** | Prepare for delivery; clerk after the baby is born |
| **Temperature 38.0°C or more** | Malaria RDT, cultures if possible, start antibiotics per protocol |
| **Gynae patient with pain + collapse** | **Urine hCG now**; ruptured ectopic until proven otherwise |

5. **If stable:** introduce yourself, get consent, arrange privacy and a chaperone, and start the full clerking.

### A. The obstetric history (in order)

1. **Introduce yourself**, confirm her identity, explain, and get **consent**. Ensure privacy and ask about a companion or chaperone.
2. **Biodata:** name, **age**, address (distance and transport), religion (blood products?), tribe/language, occupation (hers and her partner's), marital status, next of kin, **date of admission** and the referring facility.
3. **Parity and gestation:** "**G_P_+_**, number of living children, **LNMP** and **EDD**, and **GA today**." Ask about certainty of the LNMP, regular cycles, contraception before conception, and early ultrasound.
4. **Presenting complaint(s)** with duration, in her own words ("lower abdominal pain for 6 hours").
5. **History of presenting complaint:** analyse each symptom (SOCRATES for pain). Then **ask the obstetric screening questions ("PV BLOOD"):**
   - **Pain / contractions:** onset, frequency, duration, regularity.
   - **Vaginal bleeding:** amount (pads), colour, clots, painless or painful, trigger (coitus).
   - **Baby's movements:** normal, reduced or absent.
   - **Liquor draining:** when, colour (clear, meconium-stained, blood-stained), smell.
   - **Oedema and pre-eclampsia symptoms:** headache, blurred vision, epigastric pain, vomiting, swelling.
   - **Other infection:** fever, dysuria, offensive discharge, cough. Also **anaemia**: fatigue, palpitations, breathlessness.
   - **Done so far:** drugs, referral treatment such as $MgSO_4$, and **when**.
6. **History of the present pregnancy (trimester by trimester):**

- Was it **planned**? When and how was it confirmed?
- **First trimester:** bleeding, hyperemesis, fever, drugs, when ANC started.
- **ANC:** where, number of visits, **booking tests** (HIV, syphilis, HBsAg, Hb, blood group), **IPTp doses**, **Td**, iron–folate adherence, deworming, **ultrasound** findings, **BP readings**, any admissions.
- **Second and third trimesters:** quickening (when), any complications.

7. **Past obstetric history:** for **each** pregnancy in chronological order: **year, place, gestation, onset of labour (spontaneous/induced), duration, mode of delivery** (SVD, assisted, **CS and why**, incision), **complications** (PPH, retained placenta, tears, pre-eclampsia, sepsis), **baby's sex, weight and outcome** (alive, stillbirth fresh or macerated, neonatal death), and the puerperium (breastfeeding, fever). Abortions: gestation, spontaneous or induced, whether evacuation was needed, and complications. Ectopics.
8. **Gynaecological history: menarche**, cycle (length, duration, flow, pain), intermenstrual or postcoital bleeding, **contraception** (methods used, side effects, plans after delivery), **cervical cancer screening** (VIA, Pap), **STIs/PID**, infertility, fistula, gynae surgery (myomectomy means a scar).
9. **Past medical history: hypertension, diabetes, sickle cell disease, heart disease (rheumatic), asthma, epilepsy, TB, HIV** (ART regimen, VL), hepatitis, thyroid disease, **blood transfusions** (and reactions), previous admissions.
10. **Past surgical history:** abdominal or pelvic surgery, anaesthetic complications.
11. **Drug history and allergies:** current drugs (iron, ART, antihypertensives), **sulfa allergy** (SP, cotrimoxazole), penicillin allergy, herbal medicines ("**emumbwa**", local herbs to "open the way" in labour, which may contain uterotonics).
12. **Family history:** hypertension, diabetes, **twins**, sickle cell disease, congenital anomalies, TB.
13. **Social history:** marital status and **partner support**, **IPV** (ask privately: "Has anyone hurt you?"), **alcohol**, smoking, occupation and physical work, housing, source of income, **distance and transport to the facility**, money for delivery, planned birth companion.
14. **Review of systems** (brief, focused).
15. **Summarise back** to the patient to check accuracy, and ask whether she has questions or concerns ("ICE": ideas, concerns, expectations).

**Clerking template** (write it on the admission sheet in this order):

```
CLERKING SHEET (obstetric)
Date/time ........  Clerked by ........  Referred from ........
1. Biodata: name, AGE, address (distance/transport), occupation,
   marital status, next of kin, religion (blood products?)
2. G _ P _ + _   Living children _
   LNMP ........ (certain? regular? contraception?)  EDD ........
   Early scan: ..... weeks on ........   GA today: ..... weeks
3. Presenting complaint(s) + duration (her words)
4. HPC: SOCRATES + "PV BLOOD" screen (pain, bleeding, baby's
   movements, liquor, oedema/PET symptoms, other infection,
   done so far: drugs given before/at referral and TIMES)
5. Current pregnancy/ANC: booking GA, visits, HIV/syphilis/HBsAg,
   Hb, group/Rh, IPTp doses, Td, iron, scans, BP trend, admissions
6. Past obstetric (each pregnancy): year, place, GA, labour, MODE
   (CS: why? incision?), complications, baby weight/sex/outcome
7. Gynae: menarche, cycle, contraception, VIA/screening, STI/PID,
   fistula, surgery (myomectomy)
8. Medical/surgical: HTN, DM, SCD, RHD, asthma, epilepsy, TB, HIV
   (ART, VL), transfusions, operations, anaesthetic problems
9. Drugs & allergies: sulfa, penicillin; herbal medicines
10. Family: twins, HTN, DM, SCD
11. Social: support, IPV (alone), alcohol, money, transport
12. ICE: her ideas, concerns, expectations
```

**Gynae clerking additions:** LNMP (and **urine hCG for every woman of reproductive age**), menstrual pattern (PALM-COEIN for abnormal bleeding), postcoital or postmenopausal bleeding, contraception, sexual history, last cervical screening, obstetric history in brief.

### B. General examination (Hutchison order)

1. **Introduction and consent** (done above); **hand hygiene**; **position and exposure** as in section 4.
2. **General inspection from the end of the bed:** conscious, alert, in pain or distress, ill or well, **gait**, posture, colour (pallor, jaundice, cyanosis), hydration, **height** (below 150 cm) and **weight/BMI**.
3. **Vital signs: BP** (seated or left lateral, correct cuff, Korotkoff V for diastolic); **temperature, RR, $SpO_2$**.
4. **Hands: pallor** of the palms, clubbing, capillary refill, **pulse** (rate, rhythm, volume).
5. **Face: eyes**: conjunctival pallor, jaundice. **Mouth**: tongue pallor, hydration, oral thrush (HIV), dental caries. Facial puffiness (pre-eclampsia).
6. **Neck: thyroid**, lymph nodes, JVP.
7. **Breasts** (with consent and a chaperone): pregnancy changes, lumps, **nipple** shape (inverted nipples and breastfeeding).
8. **Chest:** respiratory exam. **Heart**: apex, murmurs (**a diastolic murmur is pathological**).
9. **Oedema:** legs (pitting), sacrum, **face and hands**. **Varicose veins**, **calf tenderness** (DVT).
10. **Reflexes** (patellar) and clonus when pre-eclampsia is suspected.

> CLINICAL PEARL
> The Ugandan bedside teacher's favourite opening: "**A young woman, not in obvious distress, not pale, not jaundiced, no oedema, well hydrated, afebrile, BP 120/80 mmHg, pulse 84 bpm regular.**" Pallor, jaundice, oedema, dehydration, lymphadenopathy and fever are the six **general** findings to always comment on.

### C. The obstetric abdominal examination (Hutchison order)

The order is **inspection → fundal height → lie → presentation → engagement (fifths) → auscultation of the fetal heart**. The four Leopold grips deliver lie, presentation and engagement in that order.

#### 1. Inspection

- **Size and shape:** is the abdomen distended, and is the distension compatible with pregnancy? **Longitudinal (ovoid)** means a longitudinal lie; **transverse/broad** suggests a transverse lie. An **overdistended** abdomen suggests twins or polyhydramnios.
- **Pregnancy skin changes: linea nigra**, **striae gravidarum** (pink = new, silvery white = old striae albicantes).
- **Scars: Pfannenstiel** (lower segment CS, gynae surgery), **midline** (classical CS, laparotomy), laparoscopy ports, **appendicectomy**. Ask about each.
- **Umbilicus:** flat or everted (term, polyhydramnios).
- **Fetal movements** visible.
- **Bandl's ring** (in labour) and a **distended bladder**.

**2. Superficial palpation:** tenderness (abruption, chorioamnionitis, rupture, appendicitis), **uterine consistency** (soft and relaxed, contracting, or **woody hard** in abruption), guarding. **Watch her face**, not the abdomen.

#### 3. Fundal height

- **By landmarks:** use the ulnar border of the left hand to find the upper border of the fundus, then relate it to the landmarks (section 3).
- **SFH (from 24 weeks):**
  1. Find the fundus with the left hand.

2. Place the tape measure, **numbers facing down** (to avoid bias), at the **upper border of the symphysis pubis**.
3. Run the tape along the **longitudinal axis of the uterus** (not necessarily the midline) to the top of the fundus.
4. Turn it over and read in **cm**. Expected: **SFH (cm) = gestational age (weeks) ± 2 cm** (between 24 and 36 weeks). Plot it on the SFH chart in the ANC card if available.

**4. Leopold manoeuvres: lie, presentation and engagement**

| Manoeuvre | How | What you find |
|---|---|---|
| **1. Fundal grip** | Facing the woman's head, cup **both hands around the fundus** | **What is in the fundus?** Breech: **broad, soft, irregular**, moves with the trunk. Head: **hard, round, ballotable** (moves independently of the trunk) |
| **2. Lateral (umbilical) grip** | Hands on each side of the uterus at the level of the umbilicus. **Steady one side, palpate with the other**, then swap | **Where is the fetal back?** Back: **smooth, firm, continuous resistance**. Limbs: **small irregular knobs**, kicking. Tells you the **lie** and helps the **position** (back anterior or lateral means OA; limbs anterior with the back hard to feel means OP) |
| **3. Pawlik's grip** (pelvic grip, one hand) | Facing the head, grasp the lower pole **just above the symphysis** between the **thumb and fingers of the right hand** | **What is presenting, and is it mobile?** Head: hard and round; **ballotable if not engaged**. Gently: it can be uncomfortable |
| **4. Deep pelvic grip** | **Facing her feet**, place both hands on either side of the lower uterus and **slide down into the pelvis** towards the symphysis | **Engagement in fifths** (how much head is above the brim), **attitude** (if the hand on the side of the sinciput is stopped higher than the hand on the side of the occiput, the head is flexed) |

**5. Other palpation findings**

- **Liquor volume:** fetal parts **easily felt** and the uterus "moulded" around the fetus suggests **oligohydramnios**. A **tense uterus**, a **fluid thrill** and parts that are difficult to feel suggest **polyhydramnios**.
- **Estimated fetal weight** (clinical estimate): small (below 2.5 kg), average, or big (above 4 kg). This is less accurate than ultrasound, but it helps.
- **Contractions (in labour):** frequency per 10 minutes, duration in seconds (palpate with the flat of the hand on the fundus), and strength.

**6. Auscultation of the fetal heart**

1. Locate the **fetal anterior shoulder**: between the umbilicus and the anterior superior iliac spine on the side of the back in a cephalic presentation (below the umbilicus), and **above the umbilicus in a breech**.
2. **Pinard:** place the bell firmly on the abdomen, put your ear on it and **remove your hand** (the stethoscope is held by the pressure between your ear and the abdomen).
3. **Count for a full 60 seconds** while **palpating the maternal radial pulse** to make sure you are not counting the mother's heart.
4. **Normal FHR is 110–160 bpm** and regular. In labour, auscultate **during and just after a contraction**: every **30 minutes** in the active first stage (WHO says every 15–30 minutes; the partograph records it half-hourly) and every **5 minutes** (after each contraction) in the second stage.
5. Interpret: below 110 (bradycardia) or above 160 (tachycardia: maternal fever, hypoxia, drugs), and decelerations after contractions (fetal compromise).

**Percussion** is rarely needed (polyhydramnios, or to confirm a full bladder).

**7. Complete the examination**: "I would like to complete my examination by checking the **urine** and, if indicated and not contraindicated, doing a **speculum/vaginal examination**." Cover the patient, help her sit up slowly (supine hypotension), **thank her**, **hand hygiene**, and **summarise** your findings to her and to the examiner.

### D. Examination of the external genitalia and speculum examination

1. Explain, get **consent**, **chaperone**, **empty bladder**, **dorsal (lithotomy) position** with heels together and knees apart, good light.
2. **Inspect the vulva:** discharge, bleeding, liquor, meconium, warts, ulcers, **varicosities**, scars (FGM, old tears), oedema, cord prolapse, fetal parts.
3. **Cusco's speculum:** warm and lubricated. Separate the labia with the left hand, insert **closed, at an oblique angle**, rotate to horizontal while advancing **downwards and backwards**, then **open gently** to reveal the cervix. Lock it.
4. **Look at the cervix:** colour (bluish in pregnancy), **os open or closed**, **products of conception** at the os, **bleeding from the os** or from a lesion (ectropion, polyp, **cancer**), discharge, **pooling of liquor** in the posterior fornix (ask her to cough: PROM), IUD strings.
5. Take **swabs** as needed (HVS, endocervical), and **VIA** screening if indicated (non-pregnant).
6. **Withdraw** while unlocking and rotating back, inspecting the **vaginal walls**.
7. **Sims' speculum** (left lateral position) is used for **prolapse** and **fistula** (the anterior vaginal wall).

### E. Bimanual examination (gynaecology)

1. Gloved and lubricated **index and middle fingers** of the right hand into the vagina. The **left hand is on the abdomen** above the symphysis.
2. **Cervix:** consistency, os, **cervical motion tenderness** (PID, ectopic).
3. **Uterus:** size (in weeks), **position** (anteverted or retroverted), **mobility**, surface (smooth or irregular: **fibroids**), tenderness.
4. **Adnexa:** fingers in each **lateral fornix** push up while the abdominal hand pushes down. Feel for **masses** or tenderness.
5. **Posterior fornix / pouch of Douglas:** fullness, tenderness (blood, pus), nodules (endometriosis, cancer).
6. Withdraw and **inspect the glove** (blood, discharge).

### F. Vaginal examination in labour (overview; see the partograph topic)

Assess **cervical dilatation (cm), effacement, consistency, position and application**; **membranes** (intact or ruptured, colour of the liquor); **presenting part**, **position** (sutures and fontanelles), **station**, **moulding and caput**; and the **pelvis** (sacral promontory, spines, sacral curve, subpubic angle). Do it **aseptically**, **every 4 hours** in the active phase (more often if indicated), and **not during APH** until praevia is excluded.

## 6. Findings and Interpretation

### Key examination findings and their meaning

| Finding | Think |
|---|---|
| Pallor + tachycardia | Anaemia, haemorrhage (concealed abruption, ruptured uterus, ectopic) |
| BP 140/90 mmHg or more + proteinuria after 20 weeks | Pre-eclampsia |
| Uterus **larger** than dates | Wrong dates, **twins**, **polyhydramnios**, macrosomia, **fibroids**, molar pregnancy |
| Uterus **smaller** than dates | Wrong dates, **FGR**, **oligohydramnios**, IUFD, transverse lie |
| Head at the fundus | **Breech** |
| Head in a flank, empty lower pole | **Transverse/oblique lie** |

| Finding | Think |
|---|---|
| Head **5/5 at term in a primigravida** | **CPD**, **placenta praevia**, OP, polyhydramnios, a pelvic mass, wrong dates |
| **Woody hard, tender uterus**, FHR absent | **Placental abruption** |
| Fetal parts **easily felt under the skin**, shock, contractions stopped | **Uterine rupture** |
| Three or more poles, two heads, two FHRs 10 or more bpm apart | **Twins** |
| **Pfannenstiel scar** | Previous CS until proven otherwise; ask |
| Diastolic murmur | Pathological (rheumatic heart disease) |
| Brisk reflexes with clonus | Severe pre-eclampsia / imminent eclampsia |
| Cervical motion tenderness | PID, ectopic pregnancy |
| Liquor pooling on speculum | PROM |

## How to record each part (with a normal example)

Examine in the same order every time and **document with numbers**:

| Part | Record (normal example) | Key abnormal findings to look for |
|---|---|---|
| **General** | "Alert, not in distress, not pale, not jaundiced, no oedema, well hydrated, no lymphadenopathy, afebrile" | Pallor, facial oedema, dehydration, jaundice, oral thrush |
| **Vitals** | "BP 118/72, pulse 84 regular, RR 16, T 36.8°C, $SpO_2$ 98%" | BP 140/90 or more; pulse above 100; RR above 20; $SpO_2$ below 95% |
| **Systems** | "Heart sounds normal, soft systolic flow murmur; chest clear; reflexes normal, no clonus; calves soft" | Diastolic murmur; crackles; brisk reflexes with clonus; unilateral calf swelling |
| **Abdomen: inspection** | "Distended, compatible with term pregnancy, linea nigra, no scars" | Pfannenstiel or midline scar; transverse shape; Bandl's ring; distended bladder |
| **Abdomen: palpation** | "SFH 36 cm, singleton, longitudinal lie, cephalic, LOA, 3/5 palpable, adequate liquor, uterus soft and non-tender, EFW about 3.2 kg" | SFH more than 3 cm off dates; malpresentation; woody-hard tender uterus; fetal parts under the skin |
| **Contractions** (labour) | "3 in 10 minutes, each 40 seconds, strong" | More than 5 in 10 minutes (tachysystole); weak or absent in prolonged labour |
| **Auscultation** | "FHR 140 bpm, regular, no decelerations after a contraction" | Below 110, above 160, decelerations, absent |
| **Speculum** (if indicated) | "Os closed, no active bleeding, no pooling" | Liquor pooling; bleeding from the os; lesion; products at the os |
| **VE** (labour, if no contraindication) | "Cervix 5 cm, 80% effaced, soft, well applied; membranes intact; vertex at –1, LOA, no moulding, no caput; pelvis clinically adequate" | Oedematous cervix, moulding +3, cord, meconium, brow, face |

## Red flags

**RED FLAG**

Stop the teaching exam and **act** (call for help) if you find:

- **BP 160/110 mmHg or more**, or any BP of 140/90 mmHg or more with headache, visual symptoms or epigastric pain
- **Heavy vaginal bleeding**, maternal **tachycardia above 110 bpm**, or a **shock index of 0.9 or more**
- **FHR below 110 or above 160 bpm** persistently, or **absent** FHR
- **Cord prolapse** or a visible fetal part at the introitus in a woman not yet ready to deliver
- A **tense, tender, woody** uterus
- **Fever of 38 °C or more** with an offensive discharge
- **Fits, reduced consciousness**
- Signs of **obstructed labour** (Bandl's ring, haematuria, +3 moulding)

## Bedside tests that complete the clerking

A clerking is **not complete** until the bedside tests are done and written in:

| Test | Who | Normal / action threshold |
|---|---|---|
| **Urine dipstick** | Every obstetric admission | Protein 1+ or more with BP 140/90 or more → pre-eclampsia workup; ketones (dehydration in labour); nitrites/leucocytes (UTI) |
| **Urine hCG** | **Every gynae patient of reproductive age** | Positive → think ectopic, miscarriage, molar pregnancy |
| **HemoCue Hb** | All admissions (especially labour, bleeding, pallor) | 11 g/dL or more normal; below 7 g/dL → blood ready |
| **Malaria RDT** | Fever, anaemia, or per ward protocol | Positive → treat per severity |
| **RBS** | Diabetes, glycosuria, big baby, unwell or unconscious | Hypoglycaemia or hyperglycaemia → treat and investigate |
| **HIV/syphilis dual test** | If status unknown or last negative test not in the 3rd trimester | Reactive → same-day ART; syphilis treatment; infant plan |
| **Doppler / Pinard FHR** | Every pregnant patient over about 20 weeks | 110–160 bpm |
| **Bedside clotting test** | Bleeding, abruption, IUFD, severe pre-eclampsia | No clot by 7 minutes → coagulopathy |
| **Partograph** | Active labour (cervix 4 cm or more on the modified WHO partograph) | Plot the first VE on the alert line; see the partograph topic |
| **Bishop score** | Before induction | 8 or more favourable; 6 or less → ripening |

## Laboratory and imaging

Order tests **for a reason written in your plan**, not by reflex. The common admission panels:

| Test | When | Expected / abnormal finding | How it changes management (and Ugandan availability) |
|---|---|---|---|
| **FBC** | Anaemia, bleeding, infection, pre-eclampsia, before CS | Hb 11 g/dL or more; platelets 150–400 × $10^9$/L; WCC up to about 15 (higher in labour) | Platelets below 100 → HELLP or DIC; severe anaemia → transfusion. HC IV and hospitals |

| Test | When | Expected / abnormal finding | How it changes management (and Ugandan availability) |
|---|---|---|---|
| **Blood group and cross-match** | Any bleeding, previous CS, grand multipara, anaemia, planned CS | Group and Rh; units available | Blood ready before theatre; Rh-negative → anti-D plan. HC IV and hospitals |
| **Urea, creatinine, electrolytes** | Hypertension, sepsis, vomiting, prolonged labour, oliguria | Creatinine 35–70 µmol/L in pregnancy | Above about 90 µmol/L = renal involvement. Hospitals |
| **LFTs** | Pre-eclampsia, jaundice, itching, vomiting after 16 weeks | ALT/AST normal (ALP up 2–4× is normal) | Raised ALT/AST → HELLP, AFLP, hepatitis. Hospitals |
| **Blood slide (malaria)** | Fever, positive RDT, severe illness | Parasites and density | Severe malaria → IV artesunate. HC III and above |
| **Blood and urine culture** | Fever 38.0°C or more, suspected sepsis or pyelonephritis | Organism and sensitivity | Tailor antibiotics. Regional/national referral |
| **High vaginal / endocervical swab** | Offensive discharge, PPROM (where the lab can process it), PID | Pathogens | Treat the cause (syndromic treatment where no lab). Hospital labs |
| **Obstetric ultrasound** | Uncertain dates or presentation, APH (placenta), SFH off dates, absent FHR, twins | Viability, number, presentation, placental site, liquor, growth | Changes the diagnosis and the mode of delivery. Many HC IVs, all hospitals |
| **Pelvic / transvaginal ultrasound** (gynae) | Pain or bleeding with positive hCG, pelvic mass | IUP, adnexal mass, free fluid, fibroids, ovarian cyst | Ectopic → surgery; fibroids → plan. Hospitals |
| **Serum β-hCG** | Pregnancy of unknown location, molar pregnancy | Rise of 63% or more in 48 h (viable IUP) Verify *many current guidelines use a lower minimum rise of about 35–53% in 48 h* | Ectopic management. Regional/national referral, private labs |
| **Viral load, CD4** | Women living with HIV (per national schedule) | Suppressed vs not Verify *current Uganda viral load suppression cut-off* | Adherence support; high-risk infant prophylaxis. Samples sent via the hub system |

# 7. Clinical Acumen

## Focused history: why each question matters

| Question | What it tells you |
|---|---|
| **Age** | Adolescent: CPD, pre-eclampsia, anaemia. Over 35: hypertension, GDM, aneuploidy, PPH |
| **Gravida/para and number living** | Parity risks (primigravida: pre-eclampsia, prolonged labour; grand multipara (5 or more): PPH, rupture). Previous losses; a child death or stillbirth → find out why |
| **LNMP certainty and early scan** | Everything, including preterm vs term, steroids, induction and FGR, depends on correct dating. An early scan (before 20–24 weeks) beats the LNMP; the LNMP beats fundal height |
| **Previous CS: number, indication, type of incision, interval** | TOLAC (trial of labour after CS) eligibility, **rupture risk**. A recurring indication (CPD) means a repeat CS; classical incision = no labour; interval under 18 months = higher rupture risk |

| Question | What it tells you |
|---|---|
| **Previous PPH, retained placenta** | Recurrence, so plan active management of the third stage, IV access and a hospital delivery |
| **Previous stillbirth: fresh or macerated, and why** | Fresh suggests intrapartum hypoxia (quality of care). Macerated suggests antenatal causes (syphilis, pre-eclampsia, diabetes, malaria) |
| **Previous preterm birth or second-trimester losses** | Cervical insufficiency, infection; consider progesterone or cerclage (specialist) |
| **Largest baby's weight and type of delivery** | "Proven pelvis" |
| **HIV status, ART and VL** | eMTCT plan; delivery and infant prophylaxis plan; SP contraindicated with cotrimoxazole |
| **Headache, visual symptoms, epigastric pain** | Imminent eclampsia |
| **Bleeding: painless or painful** | Painless suggests **placenta praevia**. Painful with a hard uterus suggests **abruption** |
| **Leaking liquor: colour** | Meconium suggests fetal compromise/post-term. Offensive suggests chorioamnionitis |
| **Fetal movements** | Fetal well-being; IUFD |
| **Drugs given before arrival and their times** | A woman referred with "$MgSO_4$ given" needs the **time** of the loading dose so the next dose is due 4 hours later |
| **Distance and transport** | Whether to admit her early (e.g. a previous CS living far away) |

## Thinking like a clinician

- **Always relate findings to the gestational age.** An SFH of 30 cm is normal at 30 weeks but worrying at 36. A 5/5 head is normal at 30 weeks but a warning at 40 weeks in a primigravida.
- **Compare three estimates of gestational age:** dates, early scan and fundal height. If they disagree, **trust an early scan** (under 20–24 weeks) over the LNMP, and the LNMP over fundal height. Then ask why the fundus disagrees.
- **"If the lie is not longitudinal, the delivery is not vaginal"** (transverse or oblique lie in labour means CS, unless ECV or it corrects).
- **The history gives you the diagnosis most of the time; the examination confirms it.** Form a hypothesis before you touch the abdomen, then look for evidence for and against it.
- **Pertinent negatives matter.** "No headache, no visual disturbance, no epigastric pain" in a hypertensive woman is as important as the positives.
- **SFH off dates → "Dates, Growth, Liquor"**: recheck dates, then scan for growth and liquor.
- **Escalation rule:** any red flag means **tell the midwife or doctor now**. Do not wait until the end of the teaching session.

## Common mistakes (pitfalls)

- Counting the **maternal pulse** as the FHR (always feel her radial pulse while listening).
- Measuring SFH **with the tape numbers visible** (bias), from the **umbilicus**, or with a **full bladder**.
- Mistaking a **deeply engaged breech** for a head, especially in a primigravida (the head at the fundus is the clue). Confirm with ultrasound.
- Writing "cephalic presentation" without saying **how many fifths** are palpable.
- Doing a VE in **APH**.
- Forgetting to ask the reason for a previous CS, or the **number of living children**.

## Putting it together: summary line and problem list

**Model one-line summary (the first sentence of every presentation):**

> "Mrs K.B. is a **31-year-old G6P3+2 with 2 living children**, at **33 weeks + 6 days** by a certain LNMP of 2 February 2026 (EDD 9 November 2026), admitted today from Kasangati HC IV with **painless vaginal bleeding for 3 hours**."

**Problem list** (number them, most dangerous first):

1. Antepartum haemorrhage at 33+6 weeks: praevia until proven otherwise (painless bleeding).
2. Preterm (below 34 weeks): antenatal corticosteroids and neonatal unit.
3. Poor obstetric history: macerated stillbirth at 30 weeks in 2022 (cause?) and a neonatal death.
4. Previous ectopic pregnancy (fertility counselling, contraception).
5. Hb 9.2 g/dL: moderate anaemia.

**Working diagnosis:** "APH at 33+6 weeks, **probable placenta praevia**, mother and fetus currently stable." **Plan:** no VE; two cannulae, Hb and cross-match; ultrasound for placental site; dexamethasone 6 mg IM 12-hourly × 4; admit for observation; plan delivery by CS if praevia is confirmed (timing by a senior).

> CLINICAL PEARL
> The problem list is where students gain or lose the most marks. Each problem should be a **specific, actionable** statement ("Hb 9.2 g/dL, moderate anaemia"), not a vague label ("anaemia?"). A senior should be able to read your problem list alone and know what to do today.

## Worked examples

**Worked example 1: getting the numbers right from a messy history**

Mrs K.B. tells you: "My first child was born normally in 2015 and is fine. In 2017 I lost a pregnancy at 2 months and was cleaned (MVA). In 2019 I had twins at 34 weeks; one died after 3 days. In 2022 the baby died inside me at about 7 months (30 weeks) and came out 'peeling'. In 2023 I had an operation for a pregnancy in the tube. Now I am pregnant again."

| Pregnancy | Counts as | Living child? |
|---|---|---|
| 2015 SVD, term | Para (delivery at 28 weeks or more) | Yes |
| 2017 miscarriage at 8 weeks | "+" (before 28 weeks) | – |
| 2019 twins at 34 weeks | **Para (twins count once)** | One (one neonatal death) |
| 2022 stillbirth at 30 weeks (macerated) | **Para** (28 weeks or more, alive or dead) | – |
| 2023 ectopic | "+" | – |
| Current | Gravida only | – |

**Result: G6 P3+2, 2 living children.** Macerated (not fresh) stillbirth → think antenatal causes: syphilis, pre-eclampsia, diabetes, malaria, FGR. **Gestation:** LNMP 2 February 2026 to today (27 September 2026) = 237 days = **33 weeks + 6 days**; EDD by Naegele = 2 Feb + 7 days − 3 months + 1 year = **9 November 2026**.

**Worked example 2: turning results into a decision (gynae clerking)**

A 24-year-old P1+0, LNMP 7 weeks ago, has left iliac fossa pain and spotting. Pulse 112, BP 98/60 (**shock index 1.14**). Urine hCG **positive**. Transabdominal scan: **empty uterus, free fluid** in the pouch of Douglas. HemoCue Hb 8.4 g/dL.

- Positive hCG + pain + empty uterus + free fluid = **ruptured ectopic pregnancy** until proven otherwise.

- Shock index above 0.9 and Hb 8.4 → significant intra-abdominal bleeding.
- **Decision:** resuscitate (two cannulae, cross-match 2 units, fluids), inform the senior and theatre, **emergency laparotomy (salpingectomy)**. Serum β-hCG is not needed to make this decision.

# 8. Complications and Troubleshooting

### Acting on common abnormal findings at the bedside

| Finding | Immediate action | Next steps |
|---|---|---|
| **BP 140/90 mmHg or more** | Repeat after 15 minutes of rest; urine dipstick for protein; ask about symptoms | Bloods (FBC, LFT, RFT); if **160/110 or more**: nifedipine 10 mg orally or hydralazine 5 mg IV slowly, **$MgSO_4$** if severe features (4 g IV over 5–10 minutes + 10 g IM loading) and inform a senior |
| **SFH 3 cm or more below dates** | Recheck the dates and the technique | **Ultrasound**: growth, liquor, Doppler; look for causes (hypertension, HIV, malaria, smoking) |
| **SFH 3 cm or more above dates** | Palpate for twins or polyhydramnios | Ultrasound; RBS/OGTT (GDM) |
| **Malpresentation at 36 weeks or more** | Confirm by ultrasound | Refer for **ECV** or plan **CS**; admit early in labour |
| **Head 5/5 at term in a primigravida** | Consider CPD, placenta praevia, OP | **Ultrasound** for the placenta **before** any VE; plan delivery at a CS-capable facility |
| **FHR below 110 or above 160** | Turn her to the **left lateral**, **stop oxytocin**, check maternal temperature and BP, IV fluids | Reassess the FHR; VE for cord prolapse or progress; expedite delivery if persistent |
| **Absent FHR** | Recheck with a Doppler; confirm by **ultrasound** | Manage as IUFD (see that topic) |
| **Pallor** | Check Hb, pulse, BP | Anaemia work-up; transfuse if severe and decompensated |
| **Pooling of liquor on speculum** | **No digital VE** if not in labour; check the temperature and FHR | Manage as PROM/PPROM (antibiotics, steroids under 34 weeks) |
| **Offensive discharge + fever** | Treat as **chorioamnionitis**: ampicillin 2 g IV 6-hourly + gentamicin 5 mg/kg IV daily (+ metronidazole 500 mg IV 8-hourly) | Expedite delivery |

### Emergency flow: you find an abnormal sign during bedside teaching

1. **Stop** the teaching exam and stay with the patient.
2. **Call for help** (the midwife or doctor on duty). Say clearly: "**I need help now: BP 170/115 with a headache**" (SBAR format).
3. **Position:** left lateral (aortocaval compression, and airway safety if she fits).
4. **Basic measures you can start** as a student under supervision: vital signs, oxygen if $SpO_2$ is below 95%, prepare an IV line and blood samples, prepare $MgSO_4$ or uterotonics as instructed, and document the times.
5. **Hand over** using **SBAR**: Situation, Background, Assessment, Recommendation.

### Aftercare following examination

- **Cover** the patient, help her sit up slowly (supine hypotension), and **explain the findings** in plain language.
- **Document** (date, time, findings, your name) in the file or ANC card.
- If you did a VE: note **liquor colour** and any bleeding on the glove. Check the FHR afterwards.
- **Infection prevention:** hand hygiene before and after, **sterile gloves** for VE in labour, clean the speculum per IPC (decontaminate, clean, then sterilise/HLD).

### Complications of poor technique or omissions

| Error | Consequence |
|---|---|
| Digital VE in placenta praevia | Torrential haemorrhage |
| Repeated VEs after PROM | Chorioamnionitis, neonatal sepsis |
| Prolonged supine exam after 20 weeks | Supine hypotension, fetal distress |
| Rough Pawlik's grip / deep palpation in abruption or rupture | Pain, worsened bleeding |
| Missing a scar or a previous CS | Uterine rupture from augmentation with oxytocin |
| Misidentifying the maternal pulse as the FHR | A missed IUFD or fetal distress |
| Missing breech or transverse lie at term | Undiagnosed malpresentation in labour, cord prolapse, obstructed labour |
| Not asking about HIV/syphilis status | Missed eMTCT and congenital syphilis |
| No chaperone or consent | Complaints, loss of trust, legal risk |

### Troubleshooting

| Problem | What to do |
|---|---|
| FHR heard at the same rate as the maternal pulse | Palpate the radial pulse while listening; move to the anterior shoulder; recheck with Doppler; ultrasound if doubt |
| Cannot tell head from breech | Ballotability (head moves independently of the trunk); look for the head at the fundus; say you are unsure and request an ultrasound |
| Woman uncomfortable or dizzy lying flat | Left lateral tilt at once (aortocaval compression) |
| Obese abdomen, parts hard to feel | Use SFH with caution; rely on ultrasound for presentation and growth |
| Dating estimates disagree | Early scan over LNMP over fundal height; then ask why the fundus disagrees |

## 9. Documentation and Presenting Findings

### The ward-round presentation format

1. **Summary line:** "Mrs X is a **[age]-year-old G_P_+_ ([n] living)** at **[GA] weeks by [LNMP/early scan]**, who presented **[when]** with **[main complaint(s)] of [duration]**."
2. **HPC:** key positives and **pertinent negatives** (danger symptoms absent).
3. **Pregnancy/ANC summary:** tests (HIV, syphilis, Hb, group), IPTp, Td, BP trend, scans.
4. **Past obstetric history** (brief but complete; highlight CS, PPH, stillbirth).
5. **Relevant medical, drug and social history.**

6. **Examination:** general (pallor, jaundice, oedema, BP, pulse, temperature), then abdominal (**SFH, lie, presentation, position, engagement in fifths, FHR, contractions**), then VE if done.
7. **Summary / impression:** "In summary, a … with **[diagnosis]** complicated by … ; differentials … "
8. **Plan:** investigations, treatment, monitoring, delivery plan, and **what you would do next**.

**Presenting abdominal findings in one sentence**: "Singleton, longitudinal lie, cephalic presentation, LOA, 3/5 palpable, FHR 140 bpm regular, SFH 34 cm at 35 weeks."

### Model case presentations

**ANC / antenatal ward (template):**

> "Mrs Nakato is a **29-year-old G3P2+0 (2 living)** at **34 weeks + 2 days** by LNMP, consistent with a 16-week scan. She was **admitted yesterday** from Mukono HC IV with **a headache for 2 days and a BP of 162/108 mmHg**. She has **no visual disturbance, no epigastric pain**, and reports good fetal movements. She booked at 14 weeks: **HIV-negative, VDRL non-reactive, Hb 11.8 g/dL, O Rh-positive**, 3 doses of IPTp, Td2. BP was normal until 30 weeks. Her **previous two pregnancies** were SVDs at term in 2019 and 2021 (3.1 and 3.4 kg), both alive and well, with no hypertension. There is no chronic hypertension, diabetes or renal disease. On examination she is **not pale, with pitting oedema to the knees and facial puffiness**. BP is 158/104 mmHg, pulse 88 bpm, temperature 36.8 °C, and **reflexes are brisk without clonus**. **SFH is 31 cm**; there is a **singleton fetus in a longitudinal lie, cephalic presentation, LOA, 4/5 palpable**, and the FHR is 144 bpm regular. Urine protein is 3+. **In summary, this is a multiparous woman at 34 weeks with pre-eclampsia with severe features (severe-range BP, symptoms) and possible FGR (SFH 3 cm below dates).** Plan: $MgSO_4$ (Pritchard), BP control with nifedipine, FBC/LFTs/creatinine, a growth scan, dexamethasone for lung maturity, and delivery within 24–48 hours after stabilisation."

**Labour ward (template):** summary line; labour onset and time; membranes and liquor; ANC summary; past obstetric history; exam (vitals, abdomen: contractions **x in 10 minutes lasting y seconds**, fifths, FHR); **VE: dilatation, effacement, membranes, presenting part, position, station, moulding, caput, pelvis**; impression ("**active first stage, progressing normally, partograph on the alert line**"); plan.

**Gynaecology ward (template):** summary line with **age, parity, LNMP (or menopausal status), contraception**; complaint and duration; menstrual history; sexual and contraceptive history; screening (VIA); exam including **speculum and bimanual**; impression; plan.

### Documentation

- Date, time, findings with numbers, your name and signature, in the file or ANC card.
- Write what you did **not** do and why ("VE deferred: APH, awaiting scan").
- Record the **times** of any drugs given before arrival ($MgSO_4$, antibiotics, uterotonics).

## 10. OSCE Checklist

| Step | Done well when | Critical? |
|---|---|---|
| Washes hands, introduces self, confirms identity, explains, gains **consent**, offers a **chaperone** | All five done before touching the patient | Yes |
| Ensures privacy, **empty bladder**, positions with a pillow and a slight left tilt, exposes xiphisternum to symphysis | Pubic area kept covered; asks about comfort | No |
| Comments on the **general state**: pallor, jaundice, oedema, dehydration; checks **BP and pulse** (or asks for them) | Names the six general findings | No |

| Step | Done well when | Critical? |
|---|---|---|
| **Inspection**: distension, shape, linea nigra, striae, **scars**, fetal movements, umbilicus | Specifically looks for and asks about scars | Yes |
| **Superficial palpation** for tenderness, watching her face | Eyes on her face throughout | No |
| **Fundal height** with the ulnar border of the hand | Relates it to landmarks | No |
| **SFH** with the tape (numbers down), from the upper symphysis to the fundus, in cm | Numbers hidden; along the uterine axis; reads in cm and compares with dates | Yes |
| **Fundal grip**: identifies the pole in the fundus | Distinguishes head (hard, ballotable) from breech | No |
| **Lateral grip**: identifies the back and limbs, states the **lie** | Steadies one side, palpates with the other, swaps | Yes |
| **Pawlik's grip**: identifies the presenting part, ballotability | Gentle, one hand | No |
| **Pelvic grip** facing the feet: **engagement in fifths** | States a number of fifths | Yes |
| Comments on **liquor volume** and estimated fetal size | – | No |
| **Auscultates the FHR** over the anterior shoulder for **60 s** while feeling the **maternal pulse**; states the rate | Full minute; maternal pulse palpated | Yes |
| Covers the patient, thanks her, washes hands | – | No |
| **Presents findings** concisely: "singleton, longitudinal lie, cephalic presentation, LOA, 3/5 palpable, FHR 140 bpm regular, SFH 34 cm at 35 weeks" | Every finding given with a number | Yes |
| Mentions completing the exam with **urinalysis** (and VE if indicated) | States VE contraindications where relevant | No |

# ll. Key Points

## High-yield points

- **Stability first**: ABC, vitals, shock index and FHR before a 30-minute clerking.
- **Summary line**: age, G/P/+, living children, GA **and how it was dated**, main complaint and duration.
- Para counts deliveries at **28 weeks or more** in Uganda, alive or dead; **twins count once**.
- Dating hierarchy: **early scan > LNMP > fundal height**.
- Abdominal exam order: **inspection → fundal height → lie → presentation → engagement → FHR**.
- **SFH = weeks ± 2 cm** (24–36 weeks); more than 3 cm off → ultrasound.
- Engagement = **2/5 or less** palpable.
- FHR 110–160 bpm, **60 s with the maternal pulse** palpated.
- **No digital VE in APH** until praevia is excluded; **no digital VE in PPROM** without labour.
- A **head 5/5 at term in a primigravida** needs a scan (placenta first).
- **Previous CS**: why, how many, which incision, when.
- A clerking is incomplete without urine dipstick, HemoCue and (gynae) urine hCG.

## Numbers & doses

| Item | Value |
|---|---|
| Viability threshold for parity (Uganda) | **28 weeks** (UK 24 weeks) |

| Item | Value |
|---|---|
| Twins | Count as **1** para |
| Grand multipara | **5 or more** deliveries |
| Naegele's rule | LNMP **+ 7 days – 3 months + 1 year** (+ 280 days); add (cycle – 28) days for long cycles |
| Fundal landmarks | **12** weeks symphysis; **16** halfway; **20–22** umbilicus; about **28** halfway to xiphisternum; **36** xiphisternum |
| SFH | From **24 weeks**: cm ≈ weeks **± 2 cm**; more than 3 cm off → scan |
| Engagement | **2/5 or less** palpable |
| FHR | **110–160 bpm**; count **60 seconds** with the maternal pulse |
| FHR in labour | Every **30 min** (first stage); every **5 min** (second stage) |
| VE in active labour | Every **4 hours** (more often if indicated) |
| Height risk marker | Below **150 cm** |
| Hypertension / severe | **140/90** / **160/110 mmHg** |
| Shock index (HR ÷ SBP) | **0.9 or more** abnormal; **1.4 or more** critical |
| Hb (anaemia / severe) | Below **11** / below **7 g/dL** |
| Twins on auscultation | Two FHRs **10 or more bpm apart** |
| Bishop score | **8 or more** favourable; **6 or less** needs ripening |
| Bedside clotting test | No clot by **7 minutes** → coagulopathy |
| Interpregnancy interval after CS | Under **18 months** → higher rupture risk |
| Severe hypertension | Nifedipine **10 mg** oral or hydralazine **5 mg IV** slowly |
| $MgSO_4$ loading (Pritchard) | **4 g IV** over 5–10 min + **10 g IM** |
| Dexamethasone (lung maturity) | **6 mg IM 12-hourly × 4** |
| Chorioamnionitis | Ampicillin **2 g IV 6-hourly** + gentamicin **5 mg/kg IV daily** (+ metronidazole **500 mg IV 8-hourly**) |

## Exam traps

- **"Twins at 36 weeks + a miscarriage + pregnant now = G3P2+1."** → Twins count as **one** delivery: **G3P1+1**.
- **"A stillbirth at 30 weeks goes in the '+' column."** → In Uganda anything delivered **at 28 weeks or more, alive or dead**, counts in **para**.
- **"Head 5/5 at term in a primigravida: do a VE to find out why."** → Placenta praevia is on the list, so **scan first**; never a digital VE until praevia is excluded.
- **"FHR 84 bpm: severe fetal bradycardia!"** → Did you feel the **maternal pulse** while listening? Many "bradycardias" are the mother's heart. Recheck with a Doppler and confirm by ultrasound if needed.
- **"SFH measured from the umbilicus"** or **"with the tape numbers facing up"** → Measure from the **upper border of the symphysis**, along the uterine axis, **numbers facing down**, bladder empty.
- **"Cephalic presentation"** (and nothing else) → Always add **fifths palpable**, position and FHR. "Cephalic" alone tells the senior very little.
- **"Present everything I found, in order."** → Lead with the **summary line**, then the **relevant** positives and **pertinent negatives**. A 10-minute presentation of normal findings loses marks.
- **"Previous CS noted."** → Also ask **why, how many, what incision and when**. Those four facts decide VBAC vs repeat CS.

## Questions seniors ask

### Viva questions

1. **Define gravidity and parity. What is the parity of a woman who has had twins at 36 weeks, one miscarriage at 10 weeks, and is now pregnant?** G3P1+1. Twins count as one delivery, and the miscarriage is before viability.
2. **What is the difference between lie, presentation and position?** Lie is the fetal long axis relative to the maternal axis. Presentation is the part over the inlet. Position is the denominator relative to the maternal pelvis.
3. **Where do you listen for the FHR in a cephalic LOA fetus?** Below the umbilicus on the left, over the anterior shoulder (about midway between the umbilicus and the left ASIS).
4. **What does 2/5 palpable mean?** The head is engaged: the biparietal diameter has passed the brim.
5. **Why should you not do a VE in APH?** Placenta praevia may be present, and a finger through the os can cause massive haemorrhage. Scan first, or examine in theatre (double set-up).
6. **Name three causes of a uterus large for dates.** Wrong dates, multiple pregnancy, polyhydramnios (also macrosomia, fibroids, molar pregnancy).
7. **What is the normal FHR, and what are the causes of fetal tachycardia?** 110–160 bpm. Tachycardia: maternal fever/chorioamnionitis, hypoxia, maternal tachycardia, drugs (e.g. salbutamol), prematurity, fetal anaemia.
8. **What is the purpose of the fourth Leopold manoeuvre?** To assess engagement (fifths palpable) and the attitude of the head.

### On the ward round

- **Give me the summary line.** Age, G/P/+, living children, gestation **and how it was dated**, where she came from, and the main complaint with its duration.
- **How did you date this pregnancy, and do you trust it?** State the LNMP (certain or not), any early scan and the fundal height, then say which you trust: an early scan before 20–24 weeks beats the LNMP, and the LNMP beats fundal height.
- **What is your problem list?** Numbered, most dangerous first, each problem specific and actionable (e.g. "BP 162/108 with headache: pre-eclampsia with severe features").
- **The SFH is 30 cm at 35 weeks. So what?** It is more than 3 cm below dates: think "Dates, Growth, Liquor" and arrange an ultrasound (growth, liquor, Doppler).
- **Why didn't you do a VE?** Because she has antepartum bleeding (placenta praevia not excluded), or PPROM without labour (infection risk), or she did not consent. Say so and write it down.
- **Where do you listen for the fetal heart, and how do you know it is not the mother's?** Over the fetal **anterior shoulder** (below the umbilicus on the side of the back in a cephalic presentation), for 60 seconds while palpating the **maternal radial pulse**.
- **What would complete your examination? Urinalysis**, and a speculum or VE if indicated and not contraindicated (plus the bedside tests: HemoCue Hb, and a urine hCG for a gynae patient).

## Mnemonics

### Obstetric history structure: "Please Have Past Gynae & Medical Drugs Fully Sorted"

- **P**resenting complaint (with **P**arity and **P**regnancy dates)
- **H**istory of presenting complaint (and **H**istory of current pregnancy)
- **P**ast obstetric history
- **G**ynaecological history
- **M**edical and surgical history
- **D**rugs and allergies

- **F**amily history
- **S**ocial history

**Obstetric screening questions in the HPC: "PV BLOOD"**

- **P**ain / contractions
- **V**aginal bleeding
- **B**aby's movements
- **L**iquor (leaking fluid)
- **O**edema and pre-eclampsia symptoms (headache, vision, epigastric pain)
- **O**ther infections (fever, dysuria, discharge)
- **D**one so far (treatment given, referral)

**Leopold's grips in order: "Fundus, Flanks, First pelvic, Final pelvic"**, or "**What's up? Where's the back? What's down? How far down?**"

**Findings to state after palpation: "SLiPPED FH"**

- **S**ingleton (or multiple)
- **Li**e
- **P**resentation
- **P**osition
- **E**ngagement (fifths)
- **D**escent/attitude (flexed)
- **F**undal height / SFH
- **H**eart rate (FHR)

### OSCE tips

- Talk to the patient throughout ("I'm going to press gently on your tummy; tell me if anything is sore").
- **Watch her face**, not the abdomen, during palpation.
- Use the **left hand to steady and the right hand to palpate** in the lateral grip, then swap.
- **Present in order**: inspection, palpation, auscultation. **Always give a number** (SFH in cm, fifths, FHR in bpm).
- If you are unsure of the presentation, **say so** and state that you would confirm with ultrasound. Honesty scores; guessing loses marks.

> **EXAM FAVOURITE**
> Long-case examiners at UCU and Mengo often stop you after the **summary line** and ask "**What is your diagnosis?**". Prepare a **one-sentence summary** and a **ranked differential** before you walk in. Always finish with "**and I would like to complete my examination by checking the urine and, if indicated, doing a speculum/vaginal examination**".

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SELF-TEST

# Flashcards

25 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Define gravida and para (Uganda) | **Gravida:** total pregnancies including the current one. **Para:** deliveries at or beyond **28 weeks** (the Uganda viability threshold; 24 weeks in the UK), alive or dead. **Twins = 1 para.** The '+' number is losses before viability. |
| 2 | A woman is pregnant now; she had twins at 37 weeks, a stillbirth at 32 weeks and a miscarriage at 8 weeks. Write her obstetric score. | **G4 P2+1** (the twins count as 1 delivery, the stillbirth at 32 weeks is a delivery beyond viability, the miscarriage is '+1'), with **2 living children** (the twins). |
| 3 | Lie, presentation, position: definitions | **Lie:** fetal long axis relative to the maternal long axis (longitudinal, transverse, oblique). **Presentation:** the part over the pelvic inlet (cephalic, breech, shoulder). **Position:** the denominator relative to the maternal pelvis (e.g. LOA). |
| 4 | Denominators for each presentation | Vertex: **occiput**. Breech: **sacrum**. Face: **mentum**. Shoulder: **acromion**. (Brow: frontal bone/sinciput, but it is not usually named by position.) |
| 5 | Fundal height landmarks | **12 weeks:** just above the symphysis. **16 weeks:** midway to the umbilicus. **20–22 weeks:** umbilicus. **28 weeks:** midway from the umbilicus to the xiphisternum. **36 weeks:** xiphisternum. Drops at term (lightening). |
| 6 | How do you measure the SFH correctly? | Empty bladder; find the fundus by hand; tape **numbers facing down** from the **upper border of the symphysis** along the **long axis of the uterus** to the fundus; read in cm. From 24 weeks, **SFH ≈ GA ± 2 cm**. |
| 7 | The four Leopold manoeuvres | 1. **Fundal grip:** what is at the fundus?<br>2. **Lateral grip:** where is the back (lie, position)?<br>3. **Pawlik's grip:** what is presenting, and is it ballotable?<br>4. **Pelvic grip** (facing the feet): engagement in fifths and attitude. |
| 8 | Head vs breech at the fundus | **Head:** hard, round, **ballotable** (moves independently of the trunk). **Breech:** broad, soft, irregular, moves **with** the trunk. |
| 9 | Engagement in fifths | Count the fingerbreadths of head above the brim (the whole head is about 5). **5/5** = free; **3/5** = not engaged; **2/5 or less = engaged** (the biparietal diameter has passed the brim); **0/5** = not palpable abdominally. |
| 10 | FHR auscultation technique | Find the **fetal anterior shoulder** (below the umbilicus for cephalic, above for breech), use a Pinard or Doppler, count for **60 seconds** while **feeling the maternal pulse**. Normal **110–160 bpm**. |
| 11 | Causes of a uterus large for dates | Wrong dates, **multiple pregnancy**, **polyhydramnios**, **macrosomia** (GDM), **fibroids**, **molar pregnancy**, full bladder. |
| 12 | Causes of a uterus small for dates | Wrong dates, **FGR**, **oligohydramnios** (including ruptured membranes), **IUFD**, **transverse lie**, engaged head (late). |
| 13 | Head 5/5 palpable at term in a primigravida: differentials | **CPD**, **placenta praevia** (scan before any VE), **OP position/deflexed head**, polyhydramnios, a pelvic mass (fibroid, ovarian cyst), wrong dates, and a high pelvic inclination. |

| # | Question | Answer |
|---|---|---|
| 14 | Contraindications to digital VE | **APH until placenta praevia is excluded**; **PROM/PPROM not in labour** (use a sterile speculum only, because of infection risk); no consent. |
| 15 | What do you look for on speculum examination in pregnancy? | Cervix: colour, **os open or closed**, **products at the os**, **bleeding source** (os vs ectropion, polyp, cancer), discharge, **pooling of liquor** (cough test), cord, IUD strings. |
| 16 | Bimanual exam: what do you assess? | Cervix (consistency, os, **cervical motion tenderness**); uterus (**size in weeks, position**, mobility, surface, tenderness); **adnexa** (masses, tenderness); **pouch of Douglas** (fullness, nodules). |
| 17 | Structure of a ward-round presentation | 1. **Summary line** (age, G/P, GA and how dated, complaint and duration)<br>2. HPC with key positives and negatives<br>3. ANC summary<br>4. Past obstetric, medical, drug and social history<br>5. Examination (general, abdominal, VE)<br>6. **Impression**<br>7. **Plan** |
| 18 | Six general findings to always comment on | **Pallor, jaundice, oedema, dehydration, lymphadenopathy, fever**, plus **BP and pulse**. |
| 19 | Why ask whether a previous stillbirth was fresh or macerated? | **Fresh:** died during labour, pointing to **intrapartum hypoxia** (labour care). **Macerated:** died more than 12–24 hours before birth, pointing to **antenatal causes** (syphilis, pre-eclampsia, diabetes, malaria, abruption, FGR). |
| 20 | How do you confirm twins clinically? | Uterus **large for dates**, **3 or more fetal poles** or **two heads**, **many small parts**, and **two FHRs** heard simultaneously by two examiners differing by **10 bpm or more**. Confirm by ultrasound (chorionicity). |
| 21 | First 5 minutes before clerking: which findings mean 'act first, clerk later'? | **BP 160/110 or more** or fits; **shock index 0.9 or more** or heavy bleeding; **FHR below 110, above 160 or absent**; **cord prolapse** or a fetal part at the introitus; fully dilated and pushing; fever 38.0°C or more; a gynae patient with pain + collapse (urine hCG, think ruptured ectopic). Call for help using **SBAR**. |
| 22 | History: SVD 2015 (alive), miscarriage at 8 weeks 2017, twins at 34 weeks 2019 (one NND), macerated stillbirth at 30 weeks 2022, ectopic 2023, pregnant now. Parity? | **G6 P3+2, 2 living children.** Para counts deliveries at **28 weeks or more, alive or dead** (2015, twins once, the 30-week stillbirth); '+' = miscarriage and ectopic. Macerated stillbirth → think antenatal causes (syphilis, pre-eclampsia, diabetes, malaria, FGR). |
| 23 | Which bedside tests must be done before a clerking is complete? | **Urine dipstick** (every obstetric patient), **HemoCue Hb**, **urine hCG** (every gynae patient of reproductive age), **FHR** by Doppler/Pinard, and as indicated: malaria RDT, RBS, HIV/syphilis dual test if status unknown, bedside clotting test (bleeding, IUFD, abruption), partograph in active labour. |
| 24 | The four facts you must get about every previous caesarean section | **How many**, **why** (a recurring indication such as CPD suggests repeat CS), **which incision** (classical = no labour), and **when** (interval under about 18 months raises rupture risk). Together they decide VBAC vs elective repeat CS. |
| 25 | What makes a good problem list on a ward round? | **Numbered, most dangerous first**, each item **specific and actionable** (e.g. 'APH at 33+6 weeks, praevia until proven otherwise', 'Hb 9.2 g/dL, moderate anaemia'), followed by a **working diagnosis** and a plan for each problem. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 30-year-old is pregnant for the fifth time. She had a set of twins delivered at 35 weeks (both alive), a miscarriage at 12 weeks, an ectopic pregnancy treated by salpingectomy, and a normal delivery at 39 weeks (alive). How should her obstetric score be written in Uganda?

A G5P3+1, 3 living
B G5P4+1, 3 living
C G5P2+2, 3 living
D G4P2+2, 3 living
E G5P3+2, 3 living

**2.** A 24-year-old G2P1 at 38 weeks has painless, bright red vaginal bleeding of about 300 mL. BP 110/70 mmHg, pulse 96 bpm. On abdominal exam the uterus is soft and non-tender, and the head is 5/5 palpable. The medical student wants to do a VE to assess the cervix. What is the MOST appropriate advice?

A Go ahead with a careful digital VE
B Do an amniotomy to reduce bleeding
C Do a VE only if bleeding stops
D Insert a Foley catheter to tamponade
E Do not perform a digital VE; resuscitate and arrange urgent ultrasound to locate the placenta (a gentle speculum exam is acceptable)

**3.** During Leopold's first manoeuvre on a primigravida at 37 weeks, you feel a hard, round, ballotable mass at the fundus. The lower pole feels broad and soft. Where would you expect to hear the fetal heart loudest?

A Above the umbilicus, on the side of the fetal back
B Below the umbilicus in the midline
C In the left iliac fossa
D Just above the symphysis pubis
E At the xiphisternum

**4.** A 26-year-old G1 at 36 weeks by a reliable 12-week scan has an SFH of 31 cm. BP 128/80 mmHg, no proteinuria. She is HIV-positive on TLD. What is the BEST next step?

A Reassure; SFH within 5 cm is normal
B Induce labour today
C Repeat SFH in 4 weeks
D Arrange an ultrasound for fetal biometry, liquor volume and umbilical artery Doppler
E Start low-dose aspirin

**5.** A primigravida at 40 weeks is in early labour. Abdominal palpation finds the head 2/5 palpable above the brim. Which statement is CORRECT?

A The head is not yet engaged
B The biparietal diameter has passed the pelvic brim, so the head is engaged
C The head is at station +3
D This indicates cephalopelvic disproportion
E The fetus is in a breech presentation

**6.** While auscultating with a Pinard at 32 weeks, a student counts 84 bpm and reports 'fetal bradycardia'. The mother is anxious and her radial pulse is 84 bpm. What is the MOST likely explanation and the next step?

A True fetal bradycardia; prepare for emergency CS

B Fetal heart block; refer for fetal echocardiography

C Normal for 32 weeks

D The maternal pulse has been counted; reposition over the fetal anterior shoulder while palpating the maternal pulse, or use a Doppler or ultrasound

E Give oxygen and recheck in an hour

**7.** A 32-year-old G3P2 at 37 weeks has a broad abdomen. On palpation the fundus is lower than expected and empty of poles, a hard round mass is felt in the right flank, and nothing is felt over the pelvic inlet. What is the diagnosis and the correct plan if she presents in labour?

A Transverse lie; confirm by ultrasound and deliver by caesarean section

B Breech; allow vaginal delivery

C Cephalic, unengaged; augment with oxytocin

D Twins; await spontaneous delivery

E Polyhydramnios; perform amniotomy

**8.** A 28-year-old G2P1 at 34 weeks reports a gush of clear fluid 6 hours ago. She has no contractions and is afebrile, and the FHR is 148 bpm. What is the BEST way to confirm the diagnosis?

A Digital vaginal examination to feel for membranes

B Amniocentesis

C Sterile speculum examination looking for pooling of liquor in the posterior fornix (ask her to cough)

D Wait for contractions to begin

E Abdominal X-ray

**9.** On bimanual examination of a 35-year-old P3 with menorrhagia, the uterus is enlarged to about 16 weeks' size, firm and irregular, and moves with the cervix. Urine hCG is negative. What is the MOST likely diagnosis?

A Ovarian cyst

B Intrauterine pregnancy

C Endometriosis

D Pelvic abscess

E Uterine fibroids (leiomyomas)

**10.** You are presenting on a ward round. Which is the BEST opening summary line?

A This patient came in yesterday with pain.

B Mrs A. is a 22-year-old G2P1+0 with 1 living child at 36 weeks by a 10-week scan, admitted 6 hours ago with painful regular contractions and draining clear liquor.

C Mrs A. has been in hospital for 6 hours and her BP is normal.

D Mrs A. is pregnant and wants to deliver today.

E Mrs A. lives in Mukono and is a trader.

**11.** At a routine ANC contact at 30 weeks, a G2P1 has a Pfannenstiel scar. She says her first baby was 'born through the abdomen' but does not know why. What is the MOST important next step in the history?

A Obtain details of the previous CS (indication, type of uterine incision, complications, place) and the inter-delivery interval, and request previous records

B Assume the scar is from appendicectomy

C Plan an elective repeat CS without further history

D Plan delivery at the nearest HC III

E Ignore it because the scar is healed

**12.** A 19-year-old G1 at 34 weeks has a BP of 164/112 mmHg during a bedside teaching session. She complains of a headache and 'seeing stars'. What should the student do FIRST?

A Complete the teaching examination, then report

B Repeat the BP in 4 hours

C Measure the SFH

D Tell her to rest and drink water

E Stop, call for help immediately, and place her in the left lateral position while preparing for $MgSO_4$ and antihypertensive treatment under supervision

**13.** A 27-year-old G1 at 32 weeks has an SFH of 38 cm, a tense, shiny abdomen, fetal parts that are difficult to feel, and a fluid thrill. The FHR is muffled. What is the MOST likely diagnosis and key investigation?

A Macrosomia; induction of labour

B Placental abruption; coagulation screen

C Polyhydramnios; ultrasound (AFI/deepest pool, anomalies, twins) and a glucose test for diabetes

D Fibroids; MRI

E Oligohydramnios; Doppler

**14.** In the lateral grip at 36 weeks, you feel a smooth, continuous resistance on the mother's left, slightly anterior, and small parts on the right. The head is 3/5 palpable. What is the MOST likely position?

A Right occipito-posterior

B Left sacro-anterior

C Right occipito-anterior

D Left occipito-anterior

E Direct occipito-posterior

**15.** A 45-year-old P6 presents with postcoital bleeding for 3 months. She has never had cervical screening. On speculum examination there is an irregular, friable, exophytic mass on the cervix that bleeds on contact. What is the MOST appropriate next step?

A Perform VIA and treat with cryotherapy today

B Take a punch biopsy of the lesion for histology and refer for staging (clinical exam, including a rectovaginal exam, and imaging)

C Prescribe antibiotics for cervicitis and review in 2 weeks

D Insert an IUD

E Reassure her that it is an ectropion

## Answers & explanations

**1. C** · G5P2+2, 3 living

**Gravida 5** (the current pregnancy plus 4 previous ones). **Para 2**: the twins count as **one** delivery, plus the 39-week delivery. **+2**: the miscarriage and the ectopic, both before viability. **3 living** children. A and B count the twins as 2 or omit the ectopic. D forgets the current pregnancy. E counts the twins twice.

**2. E** · Do not perform a digital VE; resuscitate and arrange urgent ultrasound to locate the placenta (a gentle speculum exam is acceptable)

**Painless bleeding + soft uterus + a high head** strongly suggests **placenta praevia**. A digital VE can provoke **massive haemorrhage**. Resuscitate, cross-match, and confirm the placental site by **ultrasound**; examine in theatre with a double set-up only if needed. A, B, C and D are unsafe.

**3. A** · Above the umbilicus, on the side of the fetal back

A **hard ballotable head at the fundus** means **breech presentation**. The fetal heart is best heard over the fetal back/anterior shoulder, which in breech is **above the umbilicus**. B, C and D fit cephalic presentations. E is too high.

**4. D** · Arrange an ultrasound for fetal biometry, liquor volume and umbilical artery Doppler

SFH is **5 cm below** reliable dates, beyond the ±2 cm range, suggesting **FGR or oligohydramnios** (HIV is a risk factor). An **ultrasound for growth, liquor and Doppler** guides further management. A is incorrect. B: do not deliver without confirming the diagnosis. C delays. E: aspirin has no role for FGR at 36 weeks.

**5. B** · The biparietal diameter has passed the pelvic brim, so the head is engaged

**2/5 or less palpable means engaged**: the widest transverse diameter has passed the inlet, corresponding to about station 0. A applies to 3/5 or more. C would be 0/5. D is the opposite (a high head suggests CPD). E: fifths refer to the head.

**6. D** · The maternal pulse has been counted; reposition over the fetal anterior shoulder while palpating the maternal pulse, or use a Doppler or ultrasound

An FHR identical to the **maternal pulse** suggests the maternal (aortic) pulse is being counted. **Always palpate the mother's pulse** while listening. Relocate the fetal anterior shoulder, or confirm with a Doppler or ultrasound. A and E act on a likely artefact. B is rare. C: a normal FHR is 110–160.

**7. A** · Transverse lie; confirm by ultrasound and deliver by caesarean section

A **broad abdomen, a low empty fundus, the head in a flank and an empty pelvis** mean **transverse lie**. In labour this is an indication for **CS** (ECV may be considered before labour at 37 weeks or more if the membranes are intact). B and C are wrong presentations. D: twins would give a large uterus with several poles. E: amniotomy with a transverse lie risks **cord prolapse**.

**8. C** · Sterile speculum examination looking for pooling of liquor in the posterior fornix (ask her to cough)

Suspected **PPROM** is confirmed with a **sterile speculum exam** (pooling, a cough test, and nitrazine or other tests if available) plus ultrasound for liquor. **Avoid a digital VE**: it introduces infection and shortens latency. B and E are not indicated. D delays management (antibiotics, steroids).

**9. E** · Uterine fibroids (leiomyomas)

An **enlarged, firm, irregular uterus** moving **with** the cervix, with menorrhagia and a negative hCG, suggests **fibroids**, which are very common in Ugandan women. A: an ovarian mass is usually separate from the uterus (a groove between them, not moving with the cervix). B: hCG is negative. C: endometriosis usually gives a fixed, tender, retroverted uterus with nodules. D: an abscess is tender with fever.

**10. B** · Mrs A. is a 22-year-old G2P1+0 with 1 living child at 36 weeks by a 10-week scan, admitted 6 hours ago with painful regular contractions and draining clear liquor.

A good summary line gives **age, gravidity/parity, living children, gestation and how it was dated, when she presented and the main complaint(s)**. It sets the context for everything else. The other options are vague, incomplete or give low-priority details first.

**11. A** · Obtain details of the previous CS (indication, type of uterine incision, complications, place) and the inter-delivery interval, and request previous records

The **indication** (recurrent, e.g. CPD, vs non-recurrent, e.g. fetal distress), **type of uterine incision** (classical vs lower segment), **complications** and **interval** determine **eligibility for a trial of labour after CS** and the rupture risk. B: a Pfannenstiel scar with 'born through the abdomen' is a CS. C: repeat CS is not automatic. D: she needs a **CS-capable facility**. E is dangerous.

**12. E** · Stop, call for help immediately, and place her in the left lateral position while preparing for $MgSO_4$ and antihypertensive treatment under supervision

Severe-range BP with **headache and visual symptoms** means **imminent eclampsia**. The student must **stop and escalate immediately** (SBAR), position her safely, and help prepare **$MgSO_4$** (4 g IV + 10 g IM loading) and an **antihypertensive** (nifedipine or hydralazine). A, B, C and D delay life-saving treatment.

**13. C** · Polyhydramnios; ultrasound (AFI/deepest pool, anomalies, twins) and a glucose test for diabetes

A large-for-dates uterus, **tense**, with **difficult-to-feel parts**, a **fluid thrill** and **muffled heart sounds** is **polyhydramnios**. Ultrasound measures the liquor and looks for **fetal anomalies** (e.g. oesophageal atresia, anencephaly) and **twins**. Screen for **GDM**. A: a big baby has easily felt parts. B: abruption gives pain, bleeding and a woody uterus. D is less likely. E is the opposite.

**14. D** · Left occipito-anterior

In a cephalic presentation (head palpable in the pelvis) the **back** marks the side of the **occiput**. A back on the **left and anterior** means **LOA**. A would have the back on the right and posterior (limbs anterior, back hard to feel). B is a breech position. C would have the back on the right. E would have limbs felt anteriorly on both sides.

**15. B** · Take a punch biopsy of the lesion for histology and refer for staging (clinical exam, including a rectovaginal exam, and imaging)

A **visible, friable, exophytic lesion** is suspicious for **invasive cervical cancer**. VIA and cryotherapy (A) are for screening **precancerous** lesions, not visible tumours. **Biopsy** for histology and **FIGO staging** (clinical exam including rectovaginal, and imaging) are needed, with referral (e.g. to the Uganda Cancer Institute). C and E delay the diagnosis. D is inappropriate.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Your first bedside presentation

At Mengo Hospital antenatal ward, Dr Mutamba asks you to clerk and present Mrs Namusoke, a 31-year-old admitted this morning with 'pain in the lower abdomen'.

**1. What must you establish in the first minute of the history?**

> **Age, gravidity and parity** (and living children), **LNMP/EDD** (and any early scan) to get the **gestational age**, and the **presenting complaint with its duration**. She is G3P2+0 (2 living), LNMP gives 35 weeks + 1 day, consistent with an 18-week scan, and she has had lower abdominal pains for 8 hours.

**2. Which screening questions do you ask in the HPC, and why?**

> **'PV BLOOD'**: **P**ain (regular contractions? preterm labour), **V**aginal bleeding (abruption, praevia), **B**aby's movements, **L**iquor draining (PPROM), **O**edema and pre-eclampsia symptoms, **O**ther infections (fever, dysuria: UTI can trigger preterm labour), **D**one so far. She reports **dysuria and frequency** for 3 days, **regular tightenings** every 10 minutes, no bleeding, no leaking and good movements.

**3. Summarise her ANC and past obstetric history as you would present it.**

> 'She booked at 16 weeks: **HIV-negative, syphilis non-reactive, Hb 11.4 g/dL, B Rh-positive**, 3 doses of IPTp, Td2. BP normal throughout. In 2018 she had an SVD at term of a 3.2 kg girl, alive and well; in 2021 an SVD at **33 weeks** of a 1.9 kg boy, who spent 2 weeks in the NICU and is alive and well. There is no significant medical history and no allergies.' A **previous preterm birth** is a major risk factor for recurrence.

**4. Examination: T 38.1 °C, pulse 102 bpm, BP 118/72 mmHg, no pallor. Abdomen: SFH 34 cm, singleton, longitudinal lie, cephalic, 4/5 palpable, FHR 156 bpm, contractions 2 in 10 minutes lasting 25 seconds, right renal angle tenderness. What is your summary and differential?**

> **'In summary, a 31-year-old G3P2+0 at 35 weeks with a previous preterm birth, presenting with threatened preterm labour, probably secondary to acute pyelonephritis.'** Differentials: **chorioamnionitis** (fever, tachycardia, fetal tachycardia; check for offensive liquor and uterine tenderness), **malaria**, established preterm labour, placental abruption (no bleeding, soft uterus, less likely).

**5. What is your plan?**

> **Investigations:** urinalysis and **urine culture**, FBC, **malaria RDT**, renal function, and a **sterile speculum** (cervix and liquor) before any digital VE; then a VE if the membranes are intact to assess dilatation. **Treatment:** IV fluids, **ceftriaxone 1 g IV daily** (or per local protocol) for pyelonephritis, paracetamol for fever. At 35 weeks, antenatal steroids are generally **not** given routinely (WHO recommends them from 24 to 34 weeks). Monitor contractions and FHR, and prepare the neonatal team.

**Take-away:** Structure beats length: summary line, focused HPC with 'PV BLOOD', ANC and past obstetric history, targeted exam, a one-sentence summary, then a ranked differential and plan.

## Case 2: The abdomen that does not match the dates

Akello, a 23-year-old G2P1 at 30 weeks by LNMP (uncertain; she was breastfeeding when she conceived), attends a HC IV ANC clinic. The midwife says 'her tummy is too big'.

**1. What history do you take to explain a large-for-dates uterus?**

> **Dating**: LNMP certainty (unreliable here: breastfeeding, lactational amenorrhoea), any early scan, quickening date. **Family history of twins**; ovulation induction. **Symptoms of GDM** (thirst, polyuria), previous big babies. Breathlessness or discomfort (polyhydramnios). Previous fibroids.

**2. What examination findings would distinguish twins, polyhydramnios and a large singleton?**

> **Twins:** many small parts, **3 or more poles**, two heads, two FHRs 10 or more bpm apart. **Polyhydramnios:** tense, shiny abdomen, **fluid thrill**, parts hard to feel, muffled FHR. **Macrosomia:** a big, easily palpable single fetus. **Wrong dates:** a normal-feeling pregnancy that is simply further on.

**3. SFH is 37 cm. You feel two hard round masses, one at the fundus on the left and one in the right iliac fossa, and hear FHRs of 136 and 152 bpm. Present your findings.**

> 'On abdominal examination the abdomen is distended more than expected for dates, with no scars. The **SFH is 37 cm**, 7 cm more than dates. I palpated **two fetal poles consistent with heads**: one at the fundus and one in the right lower quadrant, with **multiple small parts**. I heard **two separate FHRs of 136 and 152 bpm**. **Impression: a multiple (twin) pregnancy**, the first twin probably cephalic and the second breech, with uncertain dates.'

**4. What investigations and management follow?**

> **Ultrasound**: confirm twins, **chorionicity** (best before 14 weeks; now less reliable), presentations, growth, liquor, placental site. **Hb** (twins raise the risk of anaemia, so give therapeutic iron–folate), **BP and urine** at every visit (a higher pre-eclampsia risk), and more frequent ANC. **Refer to hospital** for delivery (twins are high-risk: PPH, malpresentation of twin 2, cord prolapse, preterm birth). Plan delivery at about **37 weeks** for DCDA twins (earlier for monochorionic).

**5. What counselling do you give?**

> Signs of **preterm labour**, danger signs, a **birth plan at a hospital** with a neonatal unit, identifying a **blood donor**, nutrition (an extra meal), rest, and postpartum family planning.

> **Take-away:** Large for dates means working through dates, twins, liquor, a big baby and fibroids. The hands and a Pinard can diagnose twins, but ultrasound confirms them and assesses chorionicity.

## Case 3: The gynaecology clerking

Nalwoga, a 28-year-old P1+1, presents to Mengo gynaecology clinic with lower abdominal pain and an offensive vaginal discharge for 1 week. Her LNMP was 3 weeks ago.

**1. What gynaecological history do you take, and why?**

> **Pain** (SOCRATES), **discharge** (colour, odour, itch), **fever**, **dyspareunia**, **menstrual history** (LNMP, cycle, **intermenstrual or postcoital bleeding**), **sexual history** (partners, new partner, condom use, partner symptoms), **contraception** (IUD insertion recently?), **previous STIs/PID**, recent **abortion or delivery** (her '+1': when, and was it induced? septic abortion), **HIV status**, and **cervical screening**.

**2. What must you exclude first, before anything else?**

> **Pregnancy**: a **urine hCG** (an ectopic can mimic PID). It is negative.

**3. Examination: T 38.2 °C, bilateral lower abdominal tenderness without peritonism. Speculum: mucopurulent discharge from the os. Bimanual: cervical motion tenderness and bilateral adnexal tenderness, no mass. What is the diagnosis?**

> **Pelvic inflammatory disease** (the minimum criteria are **cervical motion, uterine or adnexal tenderness**, supported by fever and mucopurulent cervical discharge). Differentials: ectopic (hCG negative), appendicitis, tubo-ovarian abscess (no mass), UTI, ovarian cyst accident.

**4. How do you manage her (outpatient, Uganda)?**

> Per the Uganda Clinical Guidelines syndromic approach: **ceftriaxone 1 g IM stat** (some regimens 250–500 mg) **+ doxycycline 100 mg orally 12-hourly for 14 days + metronidazole 400–500 mg orally 12-hourly for 14 days**. Analgesia. **Treat the partner**, advise abstinence until both are treated, offer **HIV and syphilis testing**, and provide condoms. **Admit** if she has a tubo-ovarian abscess, is pregnant, is severely ill, is vomiting, or has not improved within 72 hours.

**5. What follow-up and counselling are needed?**

> Review in **72 hours** (improvement expected). Explain the risks of **infertility, ectopic pregnancy and chronic pelvic pain** with recurrent PID. Offer cervical cancer screening (**VIA**) once the infection has settled, and contraception.

**Take-away:** In gynaecology, always exclude pregnancy first. The speculum and bimanual exam give the diagnosis, and the partner must be treated too.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.